

Day 1 — Discover the 3×3

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: build the unique 3×3 magic square with 1–9 by trial.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Primary source (today’s reading)

From *Nārāyaṇa Paṇḍita* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The challenge.** “*Place the numbers 1 through 9 in a 3×3 grid so that every row, every column, and both main diagonals sum to the same number.*” Hand out blank grids. 8 minutes of trial.

2. **Most students will find the same answer.**

2	7	6
9	5	1
4	3	8

(Or one of its 7 symmetric variants — rotations and reflections.)

3. **The magic constant.** Each row, column, and diagonal sums to **15**. Verify on the board.
4. **Why 15?** Sum of $1+2+\dots+9 = 45$. Divided across 3 rows = 15. The magic constant is forced by the digit sum.

5. **Source.** This 3×3 square appears in *Gaṇita-kaumudī* of Nārāyaṇa Paṇḍita (1356 CE), but is also found in the Chinese *Lo Shu* tradition (~5th–10th c. BCE attestation, legends earlier). The 3×3 is one of the most studied combinatorial objects in mathematics.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Discover the 3×3 .

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).